the Laplace transform, this is called the **system's transfer function**, and designate it by $H[z]$. Here is what we find:

EQUATION 33-3
Transfer function in polynomial form. The recursion coefficients are directly identifiable in this relation.

$$H[z] = \frac{a_0 + a_1 z^{-1} + a_2 z^{-2} + a_3 z^{-3} + \cdots}{1 - b_1 z^{-1} - b_2 z^{-2} - b_3 z^{-3} - \cdots}$$

This is one of two ways that the transfer function can be written. This form is important because it directly contains the recursion coefficients. For example, suppose we know the recursion coefficients of a digital filter, such as might be provided from a design table:

$a_0 = 0.389$	
$a_1 = -1.558$	$b_1 = 2.161$
$a_2 = 2.338$	$b_2 = -2.033$
$a_3 = -1.558$	$b_3 = 0.878$
$a_4 = 0.389$	$b_4 = -0.161$

Without having to worry about nasty complex algebra, we can directly write down the system's transfer function:

$$H[z] = \frac{0.389 - 1.558 z^{-1} + 2.338 z^{-2} - 1.558 z^{-3} + 0.389 z^{-4}}{1 - 2.161 z^{-1} + 2.033 z^{-2} - 0.878 z^{-3} + 0.161 z^{-4}}$$

Notice that the "b" coefficients enter the transfer function with a *negative* sign in front of them. Alternatively, some authors write this equation using additions, but change the sign of all the "b" coefficients. Here's the problem. If you are given a set of recursion coefficients (such as from a table or filter design program), there is a 50-50 chance that the "b" coefficients will have the opposite sign from what you expect. If you don't catch this discrepancy, the filter will be grossly unstable.

Equation 33-3 expresses the transfer function using *negative* powers of z, such as: z^{-1}, z^{-2}, z^{-3}, etc. After an actual set of recursion coefficients have been plugged in, we can convert the transfer function into a more conventional form that uses *positive* powers: i.e., $z, z^2, z^3, \cdots$. By multiplying both the numerator and denominator of our example by z^4, we obtain:

$$H[z] = \frac{0.389 z^4 - 1.558 z^3 + 2.338 z^2 - 1.558 z + 0.389}{z^4 - 2.161 z^3 + 2.033 z^2 - 0.878 z + 0.161}$$